

Released-Materials Reproduction, Study 1 (or1-)

Running the Anthropic Jacobian-lens battery on Qwen 3.6 27B and OLMo 3.1 32B Instruct development/methods tier — prompt-exact where released, method-reconstructed where specified, gated where not identified

J-space campaign — official-repro sideline

branch `interp-jspace.official-repro.1` — upstream `anthropics/jacobian-lens @ 581d3986`

2026-08-08/09 (OR1-A Qwen block · OR1-B OLMo fit · OR1-C OLMo lane)

Abstract

We run the released Jacobian-lens evaluations (6 sets) and experiment prompt sets (11 sets) from the public companion release of “Verbalizable Representations Form a Global Workspace in Language Models” under one audited harness on two open-weight models, and separate prompt, lens, intervention, capability, and model effects. This report covers the Qwen lane: the campaign-pinned checkpoint with the published $n=1000$ lens — the study’s mechanical instrument anchor (OR-Q1) and its first generalization surface (OR-Q2). Fidelity classes and gate states are carried on every cell; gated cells are missing, never zero; no confirmatory language is licensed anywhere in Study 1.

1 Provenance and instrument identity

Upstream release	<code>anthropics/jacobian-lens @ 581d3986</code> (11 experiments + 6 evals, hash-pinned)
Qwen checkpoint	<code>Qwen/Qwen3.6-27B @ 6a9e13bd</code> (64 layers, $d=5120$, 48 GDN blocks verified)
Published lens	<code>neuronpedia/jacobian-lens @ a4114d7</code> , $n=1000$, sha256 1718c8c5... byte-verified
Readout parity	max abs. diff 0.0e+00 (hard stop threshold 10^{-4} ; exact top-20 order agreement)
$\alpha=0$ no-op	exact (0.0 logit difference across a full-position band session)
g-folding audit	min cosine 0.382 over 26 battery tokens $\times$ 14 band layers — material ; D9: paper-literal unfolded v_t stays prima
Bands	paper grid 25 layers; workspace band [24, 58]; campaign band sensitivity [20...44]

Table 1: Lane admission. The boot-probe sentinel (Fig. 1) reads the latent country mid-band and the currency near the output — the instrument behaves as the paper describes before any experiment runs.

2 The paper’s own example, on Qwen

3 Six released lens evaluations (OR-Q1)

4 Qwen causal core (OR-Q2)

4.1 Verbal report

Incident note (or1-001): the v1 numbers below score the leading-space token form at a boundary whose in-context form is bare; they *understate* the effect (e.g. the `color:Red` swap made the model’s top-1 literally Red while the scored `_Red` form sat at rank 5). A v2 rerun with min-over-forms scoring supersedes these at the next Qwen GPU slot; see `reports/INCIDENT_or1-001_vr_token_form.md`.

Release-literal rule (first ten listed candidates, skip the answer), swap-out = greedy token at the generation boundary (D1), $\alpha=1$ across the band at every prompt position. Category-equal top-1 0.0% and top-5 0.8% over 14 categories (118 executed trials; 64 trials improved the candidate’s rank, 53 worsened). Paper context

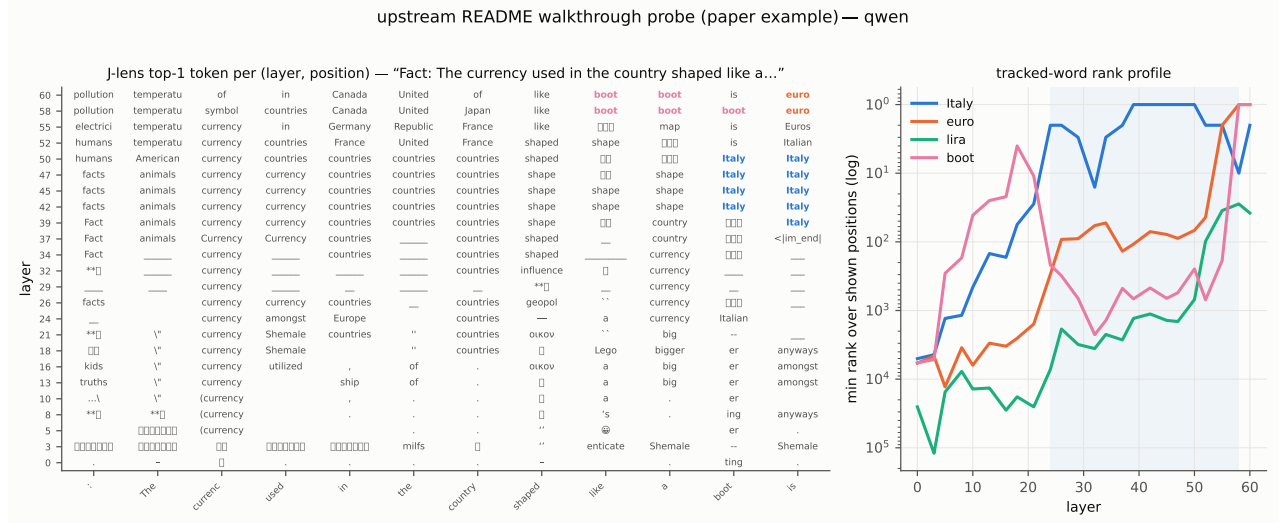


Figure 1: The upstream README walkthrough probe (“The currency used in the country shaped like a boot is”) under the published lens on Qwen 3.6 27B. Left: J-lens top-1 token per (layer, position); colored cells are tracked words. Right: tracked-word best rank by layer (shaded: workspace band). *Italy* appears as latent content mid-band and hands off to *euro*/*lira* toward the output — the paper’s slice-visualisation phenomenology, reproduced from released materials.

Set	Jacobian lens			logit lens		
	@1	@5	@20	@1	@5	@20
multihop ($n=93$)	0.244	0.399	0.641	0.173	0.327	0.542
multilingual ($n=107$)	0.237	0.384	0.580	0.184	0.327	0.497
poetry ($n=98$)	0.010	0.020	0.071	0.020	0.031	0.061
typo ($n=96$)	0.135	0.271	0.500	0.333	0.594	0.719
order-ops ($n=55$)	0.145	0.455	0.727	0.209	0.600	0.727
association ($n=102$)	0.039	0.059	0.108	0.010	0.059	0.118

Table 2: pass@ k , token-valid primary. All-source-layer sensitivity and released-denominator sensitivity live in the registered JSON.

(Claude): 88% top-5. The effect on Qwen under the paper-literal operation is *directionally present in rank movement but far below the paper’s report-rate*: Fig. 5.

4.2 Flexible generalization

All 192 ordered off-diagonal swaps; capability-conditioned primary (source and target baselines correct; baseline accuracy 51.6%); $\alpha=2$ full sensitivity. Category-equal capable top-1: 5.8% ($\alpha=1$) $\rightarrow$ 0.0% ($\alpha=2$); diagnostic all-executable: 5.6%. Paper context: 76/192 $\rightarrow$ 101/192 (Sonnet 4.5). Per-category grids in Fig. 6.

4.3 Probe-swap, raw J-lens token arm

Fidelity `prompt_exact_representation_adapted_raw_jlens` (the paper’s own §3.3 headline arm; the official linear-probe arm is NOT-IDENTIFIED — its training corpus is absent from the release). Population: 90 released items, 7 tokenization-gated, 83 executed, 63 baseline-capable. Capable top-1 14.3% (diagnostic 14.5%); multihop 10.5% vs non-multihop 15.9%. Paper context: 60% (Claude, $n=90$).

5 Worked examples: what the swaps actually do

The aggregate rates hide a bimodal reality worth seeing item by item. When the paper-literal swap works on Qwen, it works emphatically — the predicted counterfactual answer jumps hundreds of ranks to top-1:

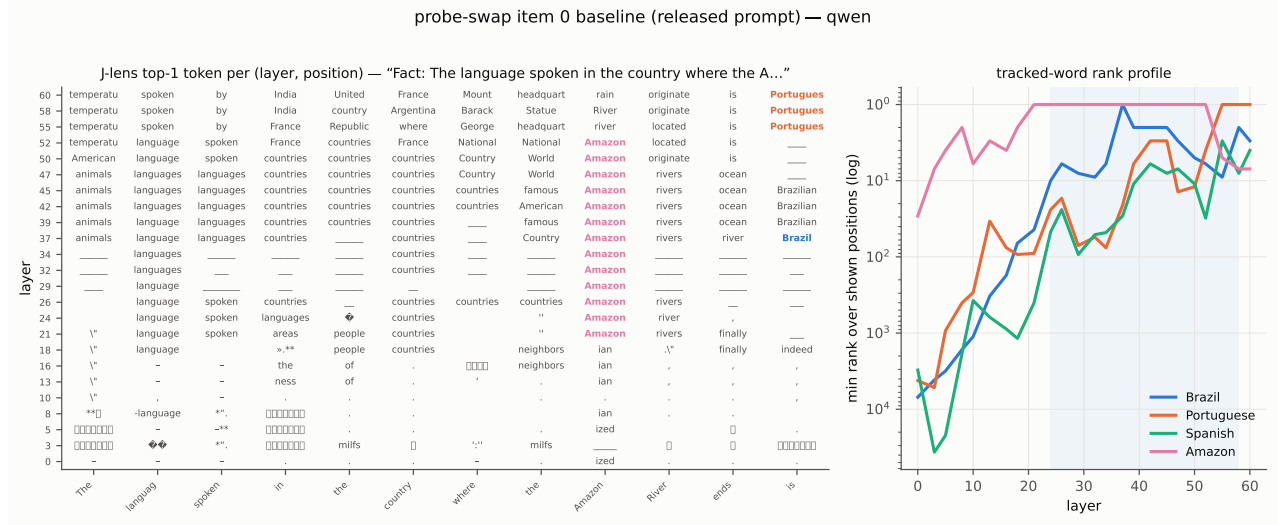


Figure 2: Released probe-swap item 0 (“the language spoken in the country where the Amazon River ends”) — the two-hop structure is visible: **Brazil** (bridge entity) carried in-band while the output side resolves toward **Portuguese**.

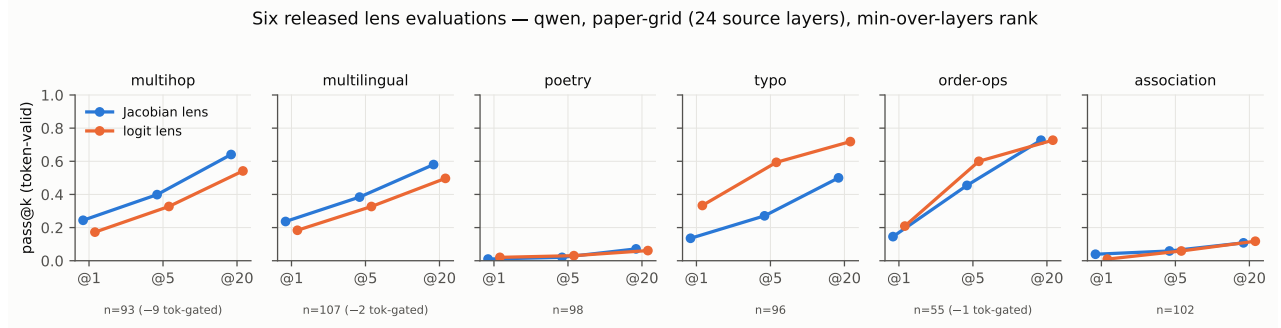


Figure 3: pass@k per released set, token-valid denominators, min-over-24-paper-grid-layers at the released readout position.

Two structural observations. First, successes cluster where the swapped-in argument has a strong, single-token, high-frequency identity (cities, countries, elements); the sparse-identity categories (months as time-words, small numbers) never flip. Second, the near-miss mass (top-5 but not top-1) plus the incident-or1-001 form effect mean the honest Qwen picture is “the mechanism visibly engages but rarely completes the flip at $\alpha=1$ ” — which makes the $\alpha=2$ reversal (0% capable) the more interesting anomaly: on Qwen the doubled patch overshoots into off-distribution territory rather than completing near-misses, unlike the paper’s Sonnet where $\alpha=2$ lifted 76/192 to 101/192.

6 OR1-B: the OLMo official-estimator fit

Timing gate (frozen before any outcome): dim_batch ladder 161/164/181/232/386 s per prompt for 16/8/4/2/1 $\Rightarrow$ dim_batch=16 at 161.4s/prompt; only the 2×125 tier fits the 18 GPU-hour ceiling $\Rightarrow$ merged $n=250$ (the addendum’s honest-prior tier). Both halves completed 125/125 with zero skips. Split-half operator audit: median symmetric relative Frobenius 8.8% across the 32 fitted layers (7.5% in-band); minus-identity cosines ≈ 1.0 (fp32-accumulation readings marginally above 1 are numerical); late layers are near-identity (identity fraction 0.93–0.97), so their low top-64 subspace overlaps reflect noise-dominated residuals, not disagreement. Readout stability: half-A/half-B eval pass@20 gaps ≤ 0.015 on every set — far inside the 0.10 margin. **No FIT-SENSITIVE classification; the merged lens is the OLMo primary.**

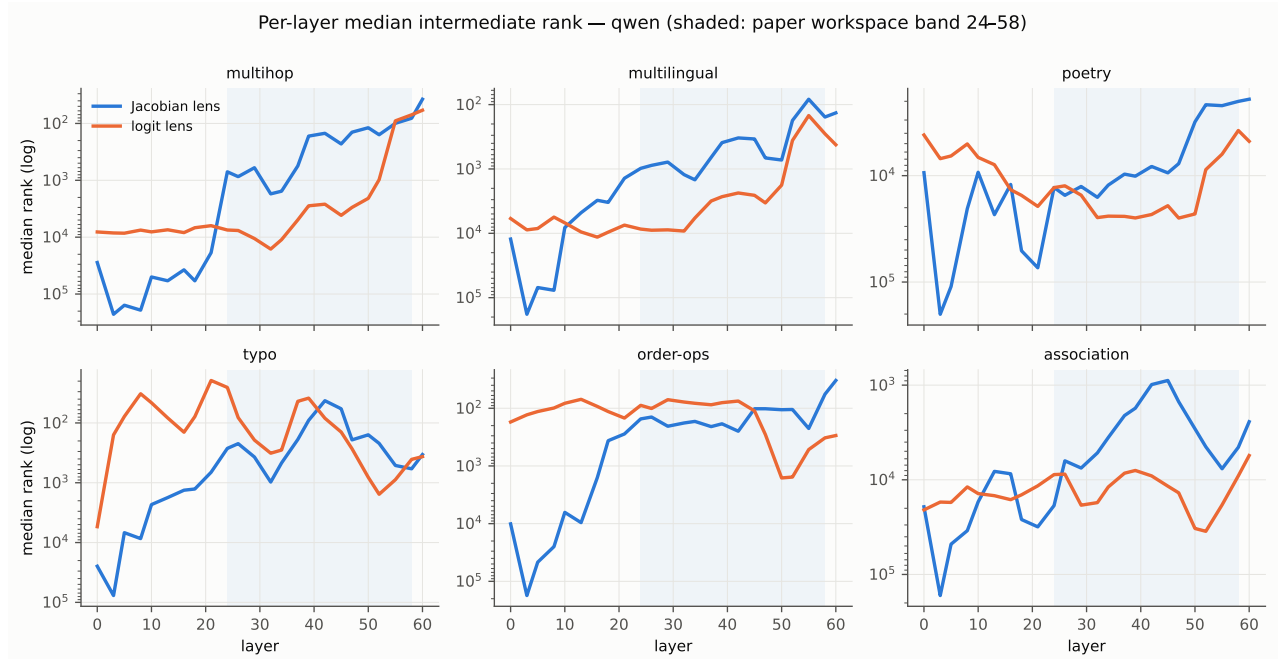


Figure 4: Median intermediate rank by layer. On the latent-content sets (multihop, multilingual) the J-lens reaches useful ranks by the band onset while the logit lens remains near rank 10^4 until $\sim L45$ — the paper’s central methods-comparison claim, visible on an open model. On typo the logit lens is stronger (the correction is surface-adjacent); order-ops numbers sit in the prompt, so the logit lens copies them early.

7 OR1-C: the OLMo lane

Admission: readout parity 0.0e+00 (exact), $\alpha=0$ no-op exact, g-folding min cosine 0.989 — *essentially immaterial*, in sharp contrast to Qwen’s 0.382 (D9 addendum): the campaign’s g-folded dictionaries and the paper’s unfolded basis are nearly the same object on OLMo but materially different objects on Qwen.

7.1 OLMo causal core (v2 scoring throughout)

Verbal report: category-equal top-1 4.8%, top-5 25.5% (118 executed; 88 improved / 27 worsened). This is an order of magnitude above Qwen’s v1 numbers (v2 rerun pending) though far below the paper’s 88% (Claude). **Flexible generalization:** capable top-1 5.2% ($\alpha=1$) $\rightarrow$ 0.0% ($\alpha=2$) — numerically indistinguishable from Qwen (5.8%/0.0%), *including the $\alpha=2$ reversal*: the over-strength overshoot is a cross-model regularity of the paper-literal swap on open models, whereas Sonnet 4.5 improved 76/192 $\rightarrow$ 101/192. **Probe-swap raw arm:** capable top-1 9.3% (diagnostic 10.7%; 43 capable of 90; paper 60%).

8 Interpretation under the wording router (two lanes)

Conformance and admission passed mechanically on both lanes (exact parity, exact no-ops, verified pins; fit stability inside every margin), so all results above are scientific observations about model/instrument cells, not harness states.

The two lanes **dissociate the readout and intervention properties**. On Qwen the J-lens readout advantage reproduces exactly as the paper describes (multihop/multilingual profiles), yet paper-literal swaps barely move behavior. On OLMo the readout advantage largely *disappears* (logit lens already strong in-band; near-identity J_ℓ), yet verbal report shows the strongest open-model swap effect so far (25.5% top-5). Meanwhile flexible generalization behaves identically on both lanes ($\approx 5\%$ capable at $\alpha=1$, 0% at $\alpha=2$) — the $\alpha=2$ reversal is a shared open-model regularity that inverts the paper’s Sonnet finding. The g-folding asymmetry (material on Qwen, immaterial on OLMo) marks a concrete basis-choice difference between the campaign’s instruments

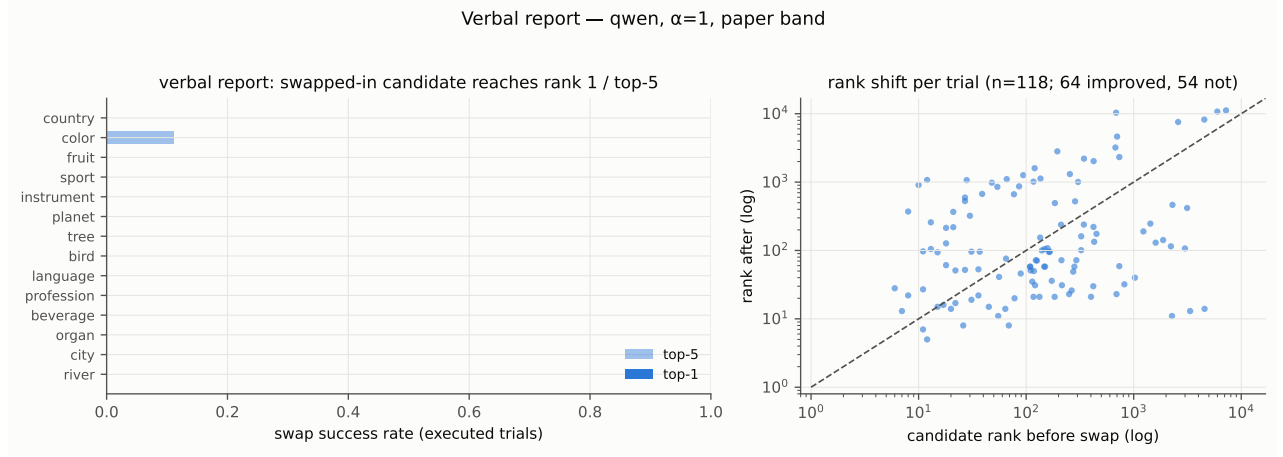


Figure 5: Verbal report on Qwen: per-category success rates (left) and per-trial rank shifts (right; below the diagonal = swap moved the candidate toward the output).

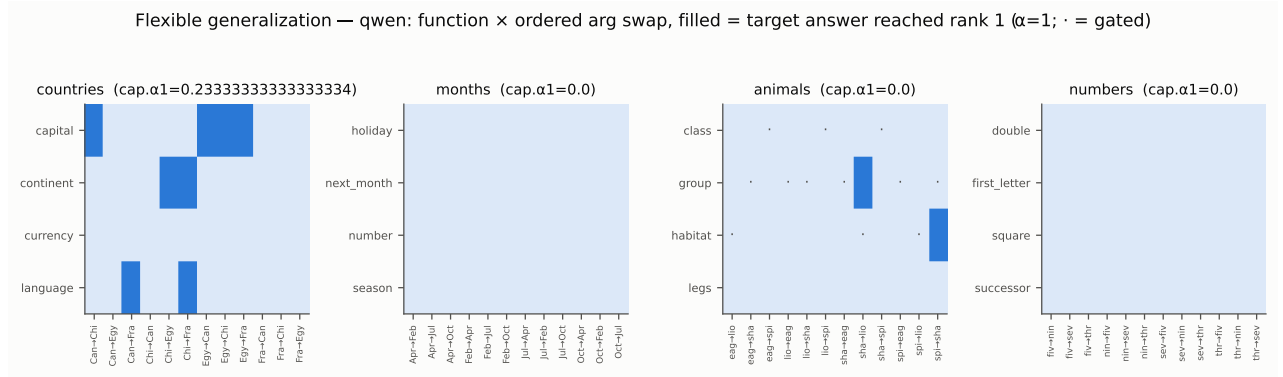


Figure 6: Function $\times$ ordered-argument-swap outcomes, $\alpha=1$ (filled = target answer reached rank 1).

and the paper’s, live on exactly the lane where interventions are weakest.

Maximum licensed sentences at this boundary: for Qwen, the plan §7.6 route “*conformance passes, core null/mixed — a valid Qwen generalization result*”; for OLMo, a property-specific version of Route A’s weaker half: verbal report is directionally present at development tier, flexible generalization is not, and no workspace-existence or -absence claim is licensed for either model. The extended battery and the instrument cross-over (separating prompt, lens, and intervention semantics — OR-Q4) complete the picture in OR1-D.

9 Population accounting

Every released row carries source-present / tokenization-valid / geometry-valid / baseline-capable / executed states in the registered raw records (`*_raw.jsonl`, plan §12 schema); the attrition waterfall figure ships with the OLMo lane so both lanes share one accounting frame. Nothing gated was encoded as zero anywhere in this report.

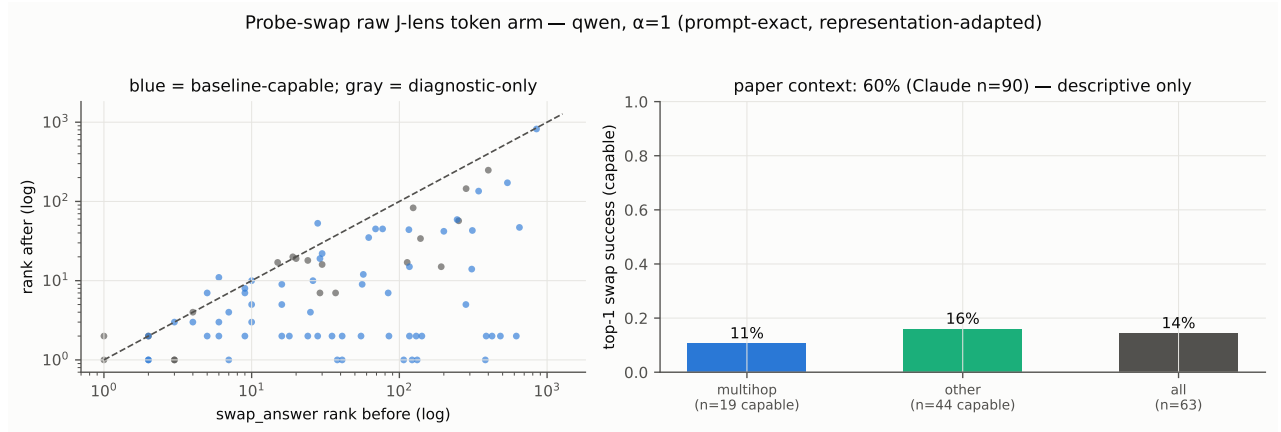


Figure 7: Probe-swap raw arm on Qwen: per-item rank movement of the predicted swapped answer (left) and capable top-1 by category group (right).

item	category	swap-answer rank before	rank after	model output after swap
ex-element-symbol-11-26	element-symbol	382	1	:
ex2-city-continent-Sydney	city-continent	132	1	South
ex2-city-capital-Munich	city-capital	122	1	Tokyo
ex2-city-language-Moscow	city-language	107	1	Russian
ex-city-capital-Lyon-Naples	city-capital	41	1	Paris
ex-city-language-Lyon-Naples	city-language	38	1	French
amazon-language	multihop	7	1	Spanish
ex-element-state-8-26	element-state	3	1	solid

Table 3: The 12 probe-swap successes (of 63 baseline-capable items), largest baseline distance first. **ex2-city-capital-Munich:** swapping the J-lens vector Munich→(released target) at every prompt position moves the predicted capital from rank 122 to the model’s literal output. One success (**rhyme-rain-neighbor**) was already rank 1 before the swap (1 such trivial cell in the count); 31 further items land in the top-5 without reaching top-1 — the modal failure is *under-strength*, not wrong-direction.

category	function	swap	rank before	output
countries	capital	Egypt→Canada	480	Ottawa
countries	capital	Canada→China	338	Beijing
countries	capital	Egypt→China	297	Beijing
animals	group	shark→lion	275	pride
countries	language	China→France	203	French
animals	habitat	spider→shark	103	ocean
countries	capital	Egypt→France	47	Paris
countries	continent	China→Egypt	14	Africa

Table 4: The 10 flexible-generalization $\alpha=1$ successes concentrate in the *countries* category (capable top-1 23.3% there): **capital Egypt→Canada** drives **Ottawa** from rank 480 to the output. Months and numbers show none — the four released categories differ enormously, exactly the per-category variance the paper’s own appendix warns about.

pass@20	merged	half A	half B	campaign	merged ₉	campaign ₉	logit
multihop	0.563	0.552	0.563	0.552	0.547	0.552	0.590
multilingual	0.511	0.505	0.519	0.512	0.463	0.467	0.560
poetry	0.092	0.092	0.092	0.092	0.071	0.071	0.082
typo	0.625	0.635	0.604	0.667	0.542	0.583	0.688
order-ops	0.582	0.582	0.582	0.609	0.555	0.545	0.736
association	0.240	0.250	0.250	0.229	0.219	0.219	0.229

Table 5: OLMo six-view eval grid (token-valid pass@20). The merged OR1 lens ($n=250$, official recipe) and the frozen campaign lens ($n=120$, campaign recipe) are concordant on the fair nine-layer intersection; halves bracket the merged lens tightly. **The logit lens matches or beats the J-lens on OLMo** — the reverse of Qwen.

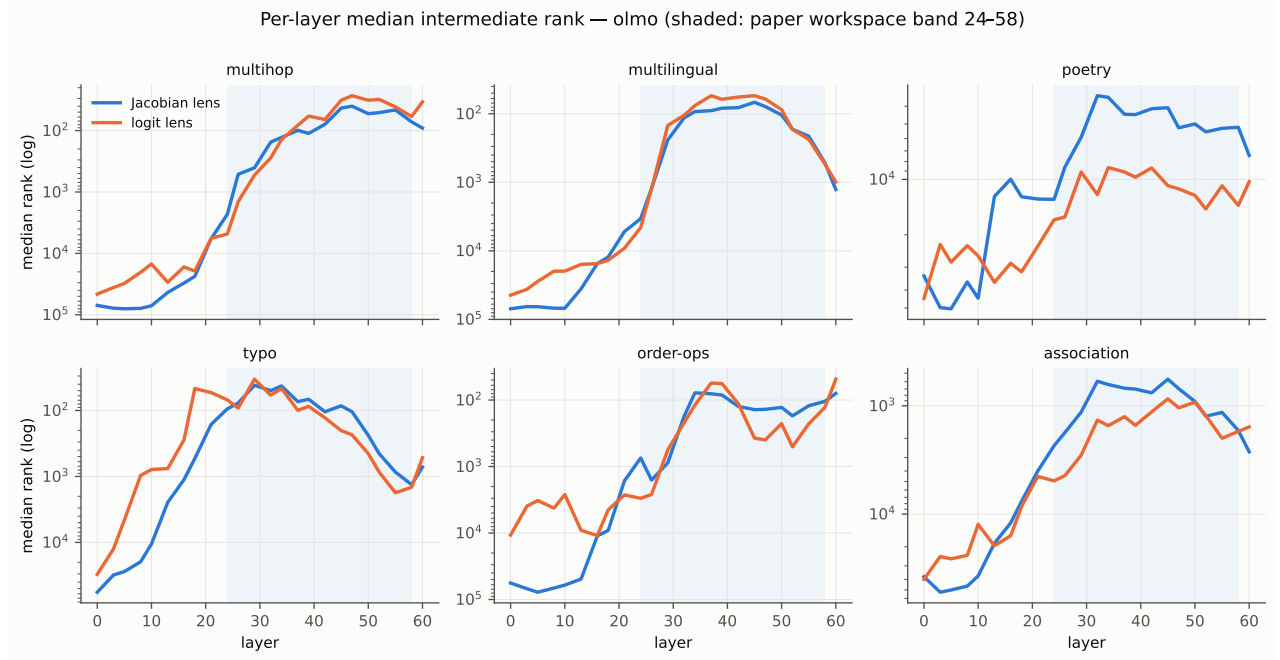


Figure 8: OLMo per-layer profiles: the two lenses nearly coincide through the band (vs. Qwen’s 1–2 order-of-magnitude J-lens advantage on latent-content sets). Consistent with the operator audit: OLMo’s in-band/late J_ℓ are close to identity, so transport adds little where the residual stream is already output-aligned. The J-lens *advantage* is itself model-dependent.

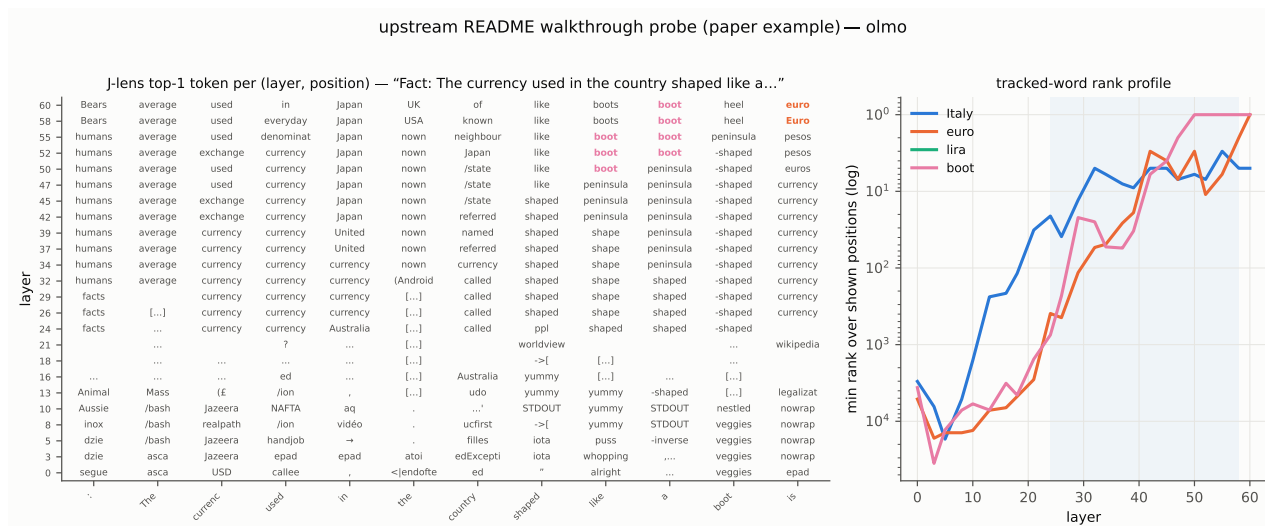


Figure 9: The boot-currency probe on OLMo with the freshly fitted $n=250$ lens: the latent country and currency handoff appear here too.

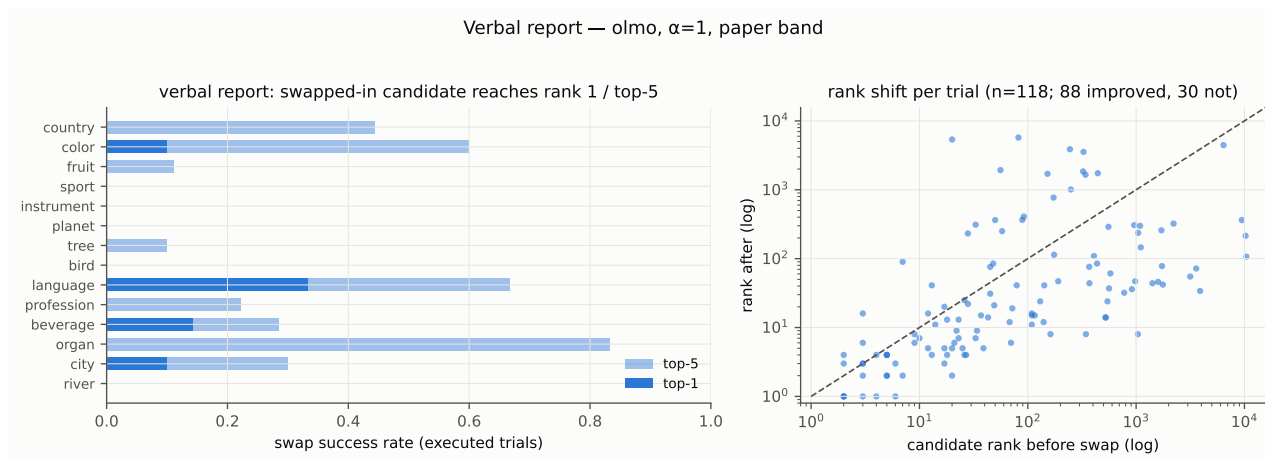


Figure 10: OLMo verbal report: per-category rates and rank shifts. The 25.5% top-5 rate is the strongest core effect either open lane has shown.